

EfficientNet-B2 Forest Density Classification Report

Model: EfficientNet-B2
Image Size: 260x260
Batch Size: 32
Optimizer: AdamW
Loss: Focal Loss (gamma=2.0)
Checkpoint Path: /kaggle/working/b2_density_best_model.pth

Dataset Split:
Train size: 4218
Validation size: 904
Test size: 905
Total size: 6027

Class Mapping:
low_density = 0
medium_density = 1
high_density = 2

Loaded Checkpoint:
Checkpoint epoch: 35
Best Validation Macro F1: 0.9126
Best Validation Accuracy: 0.9325

TEST RESULTS:
Test Loss: 0.0927
Test Accuracy: 0.9304
Test Macro F1: 0.9026

Low-Density Class:
Precision: 0.9066
Recall: 0.9510
F1-score: 0.9283

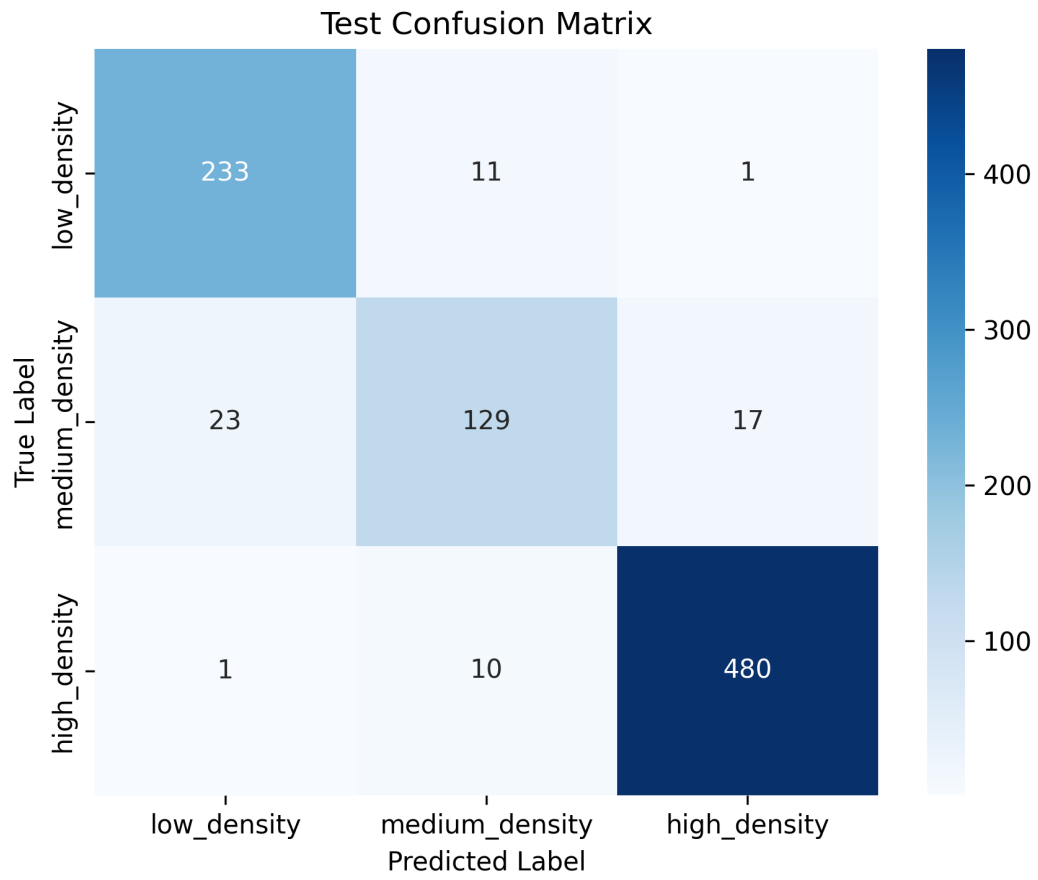
Medium-Density Class:
Precision: 0.8600
Recall: 0.7633
F1-score: 0.8088

High-Density Class:
Precision: 0.9639
Recall: 0.9776
F1-score: 0.9707

Classification Report - Test Set

class	precision	recall	f1-score	support
low_density	0.9066	0.9510	0.9283	245
medium_density	0.8600	0.7633	0.8088	169
high_density	0.9639	0.9776	0.9707	491
accuracy			0.9304	905
macro avg	0.9102	0.8973	0.9026	905
weighted avg	0.9290	0.9304	0.9290	905

Test Confusion Matrix



Model Training Report

Model: EfficientNet-B2

Image Size: 260x260

Batch Size: 32

Optimizer: AdamW

Loss: Focal Loss

Primary Metric: Validation Macro F1-score

Best Validation Result:

Stage: stage3

Epoch: 35

Val Macro F1: 0.9126

Val Accuracy: 0.9325

Val Loss: 0.1338

Best model path:

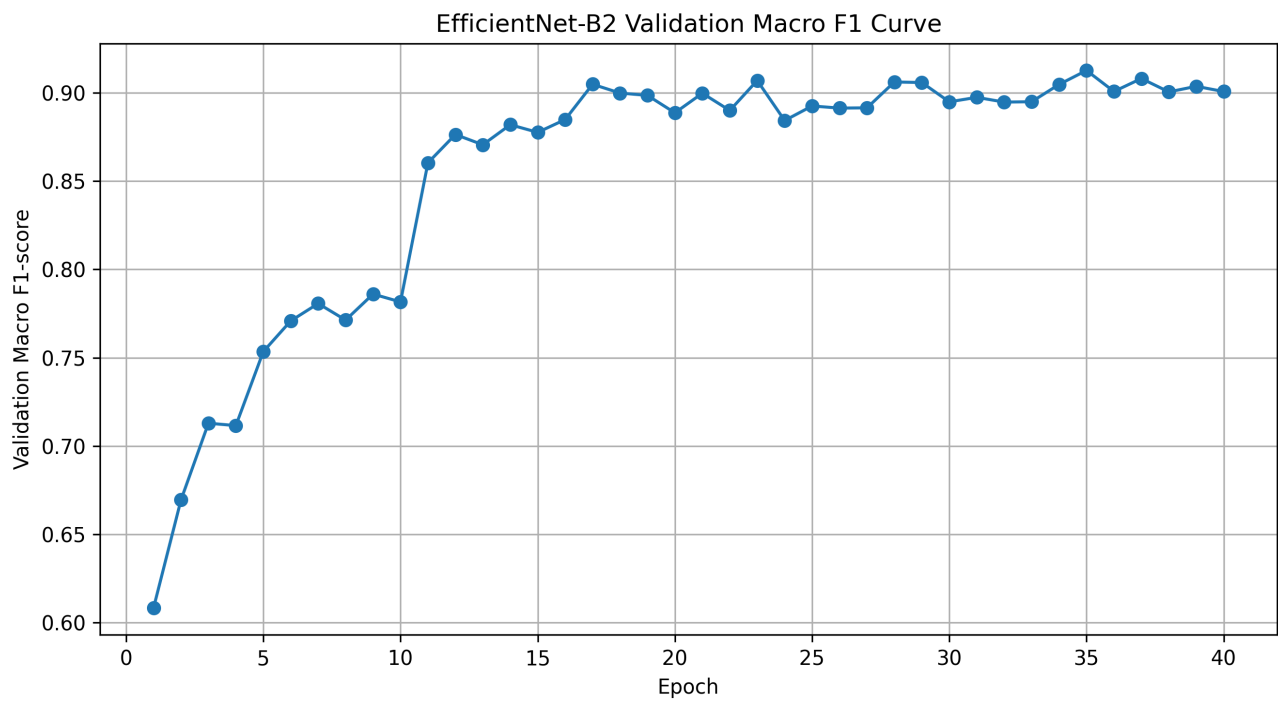
/kaggle/working/b2_density_best_model.pth

Epoch Results

stage	epoch	train_loss	val_loss	val_accuracy	val_macro_f1	time_sec
stage1	1	1.6038	1.0057	0.6659	0.6083	80.1686
stage1	2	0.9514	0.7820	0.7257	0.6695	65.3469
stage1	3	0.7678	0.6465	0.7688	0.7129	65.6188
stage1	4	0.6461	0.5880	0.7644	0.7116	66.6663
stage1	5	0.6496	0.5470	0.7987	0.7535	66.9675
stage1	6	0.5190	0.4866	0.8197	0.7708	65.3882
stage1	7	0.5182	0.4574	0.8230	0.7807	65.4826
stage1	8	0.4724	0.4386	0.8175	0.7713	66.2922
stage1	9	0.4476	0.4074	0.8285	0.7860	64.1055
stage1	10	0.3999	0.3994	0.8197	0.7816	67.6299
stage2	11	0.3693	0.2578	0.8905	0.8603	74.8200
stage2	12	0.1592	0.1889	0.9038	0.8764	74.3832
stage2	13	0.1216	0.1639	0.8993	0.8706	77.0689
stage2	14	0.0809	0.1359	0.9038	0.8819	74.2482
stage2	15	0.0513	0.1331	0.9060	0.8776	76.4195
stage2	16	0.0410	0.1231	0.9082	0.8848	77.1419
stage2	17	0.0316	0.1068	0.9248	0.9048	82.7675
stage2	18	0.0309	0.1191	0.9192	0.8998	79.2054
stage2	19	0.0265	0.1390	0.9215	0.8985	77.8060
stage2	20	0.0229	0.1262	0.9071	0.8886	78.8042
stage2	21	0.0176	0.1251	0.9204	0.8997	76.5499
stage2	22	0.0179	0.1294	0.9126	0.8900	75.2783
stage2	23	0.0147	0.1166	0.9270	0.9069	78.8445
stage2	24	0.0162	0.1612	0.9137	0.8843	75.9926
stage2	25	0.0182	0.1375	0.9137	0.8925	78.1859
stage2	26	0.0185	0.1298	0.9159	0.8913	76.5491
stage2	27	0.0140	0.1386	0.9159	0.8915	77.5503
stage2	28	0.0123	0.1235	0.9270	0.9061	74.5314
stage2	29	0.0096	0.1403	0.9259	0.9058	75.0313
stage2	30	0.0117	0.1546	0.9170	0.8948	79.0400
stage3	31	0.0150	0.1247	0.9181	0.8974	79.0722
stage3	32	0.0123	0.1229	0.9170	0.8947	78.1692
stage3	33	0.0080	0.1357	0.9192	0.8949	81.0436
stage3	34	0.0073	0.1213	0.9248	0.9047	76.2992
stage3	35	0.0095	0.1338	0.9325	0.9126	76.1061

stage	epoch	train_loss	val_loss	val_accuracy	val_macro_f1	time_sec
stage3	36	0.0103	0.1387	0.9248	0.9007	76.3168
stage3	37	0.0053	0.1439	0.9292	0.9081	75.8489
stage3	38	0.0074	0.1238	0.9248	0.9005	75.9664
stage3	39	0.0038	0.1282	0.9259	0.9036	74.2601
stage3	40	0.0063	0.1216	0.9226	0.9007	74.9220

Validation Macro F1 Over Epochs



Train Loss vs Validation Loss

